LME Metals Risk-Premia Strategies

Momentum, Carry, Value and the Combined Portfolio - Copper and Aluminium

Methodology, Research Process, and Performance
Backtest window: 2006-01-03 to 2025-12-31 (Copper), 2006-01-03 to 2026-05-19 (Aluminium)

1. Executive Summary

This document defines the four strategies presented on the dashboard - Momentum, Carry, Value, and the combined Portfolio - for both LME Copper and LME Aluminium. For each strategy it states the economic rationale, the exact signal formula, the trading assumptions, the research process used to select the winning configuration, and the realised performance over the full sample and over the trailing five years.

The central result is the diversification thesis at the heart of the framework: three economically distinct, near-uncorrelated return sources combine into a portfolio whose risk-adjusted return materially exceeds any single sleeve. For Copper the equal-weight portfolio earns a net Sharpe of 0.73 over the full sample (best single sleeve 0.62, Momentum); for Aluminium it earns 0.85 (best single sleeve 0.64). The portfolio also cuts drawdown roughly in half versus the individual signals.

A second result is that the optimal configuration is metal-specific. Copper rewards a medium-speed trend and a fast curve-momentum carry; Aluminium rewards a slower trend and a mean-reverting (z-score) carry. The dashboard auto-switches to each metal's best configuration when the metal toggle is changed, and every number in this document is reproduced live there.

Headline figures (net of 5 bps round-trip transaction costs, active-day Sharpe):

Metal	Best single sleeve	EW Portfolio (full)	EW Portfolio (trailing 5y)	Walk-forward OOS
Copper	Momentum 0.62	0.73	0.88	0.41
Aluminium	Carry 0.64	0.85	0.60	1.12

2. Common Methodology and Assumptions

All four strategies share one backtesting engine. The conventions below apply uniformly so that sleeves are directly comparable and can be combined without distortion.

2.1 Data and the continuous front-month series

- **Source.** LME daily settlement prices, 2006 to present. The futures curve provides contracts F1 through F27 (Price, Volume, Open Interest); the cash and 3-month forward file provides the Cash and 3M series used by some carry variants.
- **Continuous price.** Trading and PnL use a single back-adjusted front-month series, F1_continuous, built by stitching successive front contracts with a ratio (multiplicative) back-adjustment across each roll. This removes the artificial price jump on roll dates so that daily returns are clean and continuous.

- **Cross-metal build.** The Aluminium F1_continuous is built with the identical four-phase roll logic as Copper; its return series correlates 0.9994 with the validated Copper build, confirming the pipeline transfers cleanly across metals.
- **Signal vs PnL.** Signals are computed on raw, observable prices (F1_raw or the relevant Fk). PnL is always earned on F1_continuous. Transaction costs are charged on F1_raw, the actual traded price.

2.2 Transaction costs

Costs are modelled as a round-trip charge on each change in position: cost = |Δposition| × (bps / 10000 / 2) × F1_raw. The dashboard exposes 0, 5, 10 and 20 bps. Headline numbers in this document are net of 5 bps, with gross shown alongside. Low-turnover signals (z-score carry, slow trend) are barely affected; high-turnover signals are penalised, which is intentional.

2.3 Performance metrics

- **Active-day Sharpe.** annualised mean / standard deviation of daily returns × sqrt(252), computed over active days only (days the strategy holds a non-zero position) so that flat periods do not dilute the risk-adjusted measure.
- **Annualised return.** mean daily PnL return × 252.
- **Max drawdown.** the largest peak-to-trough decline of the cumulative return path (in cumulative return points).
- **% Flat.** share of days with no position, indicating how selective a signal is.

2.4 In-sample versus walk-forward (how robustness is tested)

Two lenses are reported throughout. The in-sample (IS) backtest uses the full history and is useful for ranking variants. The walk-forward out-of-sample (OOS) test re-applies a fixed configuration on a rolling basis - train on 1260 trading days (about five years), trade the next 252 days, roll forward - and reports the average out-of-sample Sharpe across those untouched one-year windows (the same convention shown on every dashboard tab, including the new Portfolio OOS section). A configuration is only trusted when its OOS Sharpe holds up; signals that look strong IS but collapse OOS (the Value V2 Baz-Granger reversal is the textbook example) are flagged and kept off the default path.

3. Momentum Strategy

3.1 Economic rationale

Commodity futures trend: supply and demand shocks resolve slowly, so price moves persist over weeks to months. Momentum monetises this persistence by holding the metal in the direction of its recent trend. Because the trend reverses gradually, momentum has no same-day urgency.

3.2 Definition and formula

The production signal is a dual moving-average crossover on the raw front price:

signal(t) = sign[MA_fast(F1_raw, m) − MA_slow(F1_raw, n)]

position(t) = +1 when the fast average is above the slow (uptrend), −1 when below. The position is shifted one day for execution (Same-Day convention). PnL = position × ΔF1_continuous.

Alternative families evaluated and shown on the tab: a single-scale CTA signal, the three-scale Baz-Granger CTA response (detailed below), and an equal-weight blend of three structural anchors MA(10,25)+MA(35,43)+MA(63,100).

3.2.1 Baz-Granger CTA response function (alternative trend signal)

This is the trend signal from Baz, Granger et al. (Man-AHL, "Dissecting Investment Strategies in the Cross Section and Time Series"). It is built per timescale and then blended. For each of three fast/slow exponential-moving-average pairs (S, L) = (8, 24), (16, 48), (32, 96) days:

Step 1 - raw trend: $x = EWMA\_S(price) - EWMA\_L(price)$

Step 2 - price-vol normalise: $y = x / StdDev\_63(price)$

Step 3 - standardise: $z = y / StdDev\_252(y)$

Step 4 - response function: $u = z \cdot \exp(-z^2 / 4) / 0.89$

- **x (raw trend).** the difference of a fast and a slow exponentially-weighted moving average of price; positive in an uptrend.
- **y (vol-normalised).** dividing by the 63-day rolling standard deviation of price puts the signal in volatility units, so it is comparable across price levels and regimes.
- **z (standardised).** dividing by its own 252-day (one-year) rolling standard deviation rescales the signal to roughly unit variance, so z is a standardised trend-strength score.
- **u (response).** the response u rises with z for weak-to-moderate trends, peaks near $z \approx \sqrt{2}$, then DECAYS for large $|z|$. Economically it cuts conviction when a trend is overextended (and so prone to reversal). The constant 0.89 normalises the response to roughly unit amplitude. The three per-timescale u values are averaged and the position is sign(average u).

3.3 Research process and selection

- **Parameter scan.** the dashboard scans an explicit set of known-good pairs plus a coarse grid of (fast, slow) windows, scoring each on full-sample gross Sharpe, then re-checks the leader under both timings and net of costs.
- **Metal-specific optimum.** Copper rewards a medium-speed trend, but the full-history grid optimum, MA(35,43), has only an 8-day gap (ratio 1.23) between its fast and slow windows - not a genuine multi-timescale separation, and the entire top of the 7,875-pair grid clusters in that same narrow region. The production default is therefore MA(20,45) (ratio 2.25, a real fast/slow gap and closer to conventional trend-desk practice): weaker full-sample (net 0.62 vs 0.70) but stronger over the trailing five years (net 0.62 vs 0.57) and in the walk-forward post-2022 windows (net 0.76 vs 0.49). MA(35,43) remains on the dashboard as a direct comparison, not removed. Aluminium's fast trend is weak; it wants a much slower filter, MA(60,115) (ratio 1.92, already a genuine gap - unaffected by this change), reflecting a smoother, more contango-driven price process.
- **Robustness.** MA(20,45)'s edge survives the Same-Day to Lag-1 step (gross 0.63 vs 0.62) and holds walk-forward (OOS average 0.36 net across all windows; 0.76 net in the post-2022 windows specifically), so it is genuine trend, not noise, and it has a defensible multi-timescale structure a trend desk would recognise. MA(35,43)'s walk-forward OOS average is higher overall (0.47) but weaker recently (post-2022 0.49) - the trade-off is discussed as an open question for Hartree in the Caveats.
- **Trader-convention check.** prompted by Prof. Ilia's feedback that MA(35,43) is a grid-search optimum rather than a convention a real trend desk would run. Trend desks typically layer fast,

medium and slow crossovers rather than trading one bespoke pair - which is exactly the structure of the Anchors blend above (MA(10,25)+MA(35,43)+MA(63,100), weights re-optimised each walk-forward window). Tested directly against a more textbook fast/medium/slow convention, MA(1,20)/(5,60)/(20,250), under the identical weight-optimisation methodology: the Anchors periods hold up out-of-sample (+0.43 average, +0.31 post-2022), while the faster/slower convention does not (-0.07 average, -0.34 post-2022). The three-speed structure is sound; these particular extreme periods are not. See the open question to Hartree in the Caveats section.

3.4 Assumptions

- position is ±1 (always in market, long or short); turnover is low so costs are minimal (net ≈ gross).
- Same-Day execution (signal close t, return t to t+1).
- no volatility targeting - sizing is handled at the portfolio layer.

3.5 Performance

Copper: MA(20,45) (MA(35,43) shown on the tab for comparison). Aluminium: MA(60,115). Net figures are after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.63	0.62	+16.2%	-52%	1%
Copper - trailing 5y	0.63	0.62	+13.2%	-32%	0%
Aluminium - full sample	0.48	0.48	+10.7%	-50%	2%
Aluminium - trailing 5y	0.38	0.38	+9.1%	-50%	0%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.36 (MA(20,45); the legacy MA(35,43) was +0.47 - see the trader-convention discussion in 3.3), Aluminium +0.34.

Trailing-five-year behaviour.

Copper momentum (MA(20,45)) held up through 2021-2025, with net Sharpe essentially unchanged from its full-sample figure (0.62 vs 0.62) and a shallower drawdown (-32%) than over the full history (-53%) - unlike the legacy MA(35,43), whose recent Sharpe (0.57) undershoots its own full-sample number (0.70), with a correspondingly weaker post-2022 walk-forward OOS. Aluminium's slow trend gave back some edge in the choppier recent regime (Sharpe 0.37 vs 0.48 full-sample) but stayed positive and fully invested, confirming the slower filter is the right choice for that metal.

4. Carry Strategy

4.1 Economic rationale

The shape of the futures curve carries information about physical tightness. Backwardation (front above deferred) signals scarcity and a positive expected roll return to a long; contango signals surplus. Carry trades the curve slope. Unlike momentum, the carry level is a state variable, so the timing question is genuinely strategy-specific.

4.2 Definitions and formulas

The base roll yield is the front-end slope:

roll_yield(t) = (F1 − F2) / F1

Several signal constructions are evaluated on the tab:

- **V1 level.** position = sign(roll_yield). Honest Sharpe ≈ 0.10 on Copper - kept as the baseline.
- **V4 carry-momentum (20-day).** position = sign(roll_yield − roll_yield[t−20]). Trades the 20-day change in the slope; a steepening curve (backwardation building) flags physical tightening. Best walk-forward carry on Copper.
- **V3 z-score (252-day).** position = sign((roll_yield − mean_252) / std_252). Standardises the slope over a rolling year; clean, low-turnover, and the best carry on Aluminium.
- **V2 long-dated slope.** deferred-tenor slope (Fj − Fk)/Fk across ten tenor pairs (F3-F15 … F12-F24) - shown for completeness but not selected.

Carry uses Same-Day execution: the slope is a level, and the flip day itself tends to coincide with the large directional move, so waiting an extra day forfeits the signal.

4.3 Research process and selection

- **Copper.** on Copper the carry-momentum 20-day signal is the standout - net 0.47 IS and the best walk-forward of any carry variant - because it is additive to price momentum (position correlation only +0.12) rather than a second trend signal.
- **Portfolio-leg choice differs from the tab default.** Carry-momentum-20d is the best STANDALONE carry signal on this tab, but it is not the Carry leg used inside the combined Portfolio (Section 6). Mechanically it trades the CHANGE in the roll yield - a momentum operation applied to a curve variable, not a curve LEVEL measure - which blurs the three-distinct-premia taxonomy the Portfolio is built on. The Portfolio's Carry leg is therefore Z-score-252d instead (net 0.23 full-sample on its own - weaker in isolation, but the conceptually correct 'Carry' slot). See the research note in Section 6.6 for where carry-momentum's own edge is better placed.
- **Aluminium. the 20-day carry-momentum is effectively dead on Aluminium; the 252-day z-score is the winner there (net 0.64 IS, OOS 0.89), matching its smoother curve dynamics.**

4.4 Assumptions

- position ±1; PnL on F1_continuous; costs on F1_raw.
- Same-Day execution.
- the z-score needs a 252-day warm-up; carry-momentum needs 20 days.

4.5 Performance

Copper: carry-momentum 20-day. Aluminium: z-score 252-day. Net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.52	0.47	+13.5%	-47%	0%
Copper - trailing 5y	0.72	0.65	+15.1%	-35%	0%
Aluminium - full sample	0.66	0.64	+14.1%	-50%	5%
Aluminium - trailing 5y	0.32	0.28	+7.5%	-50%	0%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.25, Aluminium +0.89.

Trailing-five-year behaviour.

Copper carry was the best-performing sleeve of the last five years (net 0.65), as repeated inventory squeezes and backwardation episodes in 2021-2024 played directly to the curve-momentum signal. Aluminium carry, very strong over the full sample, was more subdued recently (net 0.28) as the curve spent long stretches in stable contango; the z-score still added positive, low-turnover return and diversified the trend sleeve.

5. Value Strategy

5.1 Economic rationale

Value is the contrarian counterweight to momentum: when a price has stretched far from its own long-run anchor it tends to mean-revert. Value is deliberately the weakest stand-alone sleeve, but its negative correlation to momentum (about −0.2 to −0.4) makes it valuable inside the portfolio - it cushions exactly when trend reverses.

5.2 Definitions and formulas

- **V1 MA-reversion.** deviation = (Fk − MA_N) / MA_N on a deferred contract Fk. Position is +1 (buy) when the deviation falls below −10%, −1 (sell) when it rises above +10%, and flat in the ±10% band. This is the production signal.
- **V2 Baz-Granger reversal.** the long-horizon reversal signal from Baz, Granger et al. Detailed below. Higher in-sample Sharpe at long lookbacks but collapses out-of-sample - retained on the tab as a cautionary alternative, never the default.

Value uses Lag-1 / Same-Day with no urgency; the signal evolves slowly. The ±10% deadband keeps the V1 strategy flat (and cost-free) for long stretches, trading only genuine dislocations.

5.2.1 Baz-Granger long-horizon reversal (V2, formula detail)

Where momentum exploits short-horizon trend persistence, the Baz-Granger "value" leg exploits the opposite effect at long horizons: prices that have run a long way tend to mean-revert. The construction is deliberately simple - the signal is the price change over a long lookback N, sign-reversed so it is contrarian:

reversal(t) = F1_raw[t − N] − F1_raw[t]

position(t) = sign(reversal(t)) = +1 if the price has FALLEN over the last N days, −1 if it has RISEN.

- **Lookback N.** N is the lookback in trading days, tested at 1, 3, 5, 7 and 10 years (252, 756, 1260, 1764, 2520 days). Longer N targets slower, more structural reversals.
- **Sign logic.** reversal > 0 means today's price is below where it was N days ago - the contract is judged "cheap" and the rule goes long (+1); reversal < 0 means it has appreciated and is judged "rich", so the rule goes short (−1). It is always in the market (no deadband, unlike V1).
- **Execution.** the position is shifted one day for execution (no look-ahead); PnL = position × ΔF1_continuous, with transaction costs on F1_raw on each flip.

The non-monotonic behaviour by lookback (3yr and 10yr work, 5yr is a trap with negative Sharpe) and the out-of-sample collapse are exactly why V2 is shown only as a cautionary alternative and V1 MA-reversion is the production value signal.

5.3 Research process and selection

- **Reference point.** the NGL energy risk-premia paper uses F12 with a five-year anchor. That contract is optimal for energy but not for Copper.
- **Metal-specific optimum.** sweeping the contract (F1, F5, F8, F12) and lookback (1, 3, 5, 7, 10 years), Copper's empirical optimum is F8 at a five-year anchor; Aluminium's is F12 at five years (closer to the energy convention).
- **IS vs OOS. the V1 MA-reversion is robust out-of-sample (Copper OOS 0.77), whereas the V2 reversal's in-sample strength is a 2020-21 dislocation mirage that disappears OOS. V1 is therefore the default; V2 is shown only to demonstrate why it is not used.**

5.4 Assumptions

- three-state position (+1 / 0 / −1) with a ±10% deadband; large flat share by design.
- five-year (1260-day) moving-average anchor, minimum 630-day warm-up.
- PnL on F1_continuous; costs on F1_raw; low turnover.

5.5 Performance

Copper: V1 F8, 5-year anchor. Aluminium: V1 F12, 5-year anchor. Net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.33	0.32	+4.5%	-67%	46%
Copper - trailing 5y	0.03	0.02	+0.5%	-45%	26%
Aluminium - full sample	0.24	0.23	+2.8%	-71%	51%
Aluminium - trailing 5y	0.24	0.24	+3.7%	-55%	40%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.77, Aluminium +0.73. The per-window average flatters value because it trades selectively (flat ~40-50% of days); its full-sample (+0.32) and trailing-5y Sharpes are the more representative stand-alone figures.

Trailing-five-year behaviour.

Value is regime-conditional: most of its full-sample edge came from the 2020-2021 dislocation. Over the trailing five years Copper value was essentially flat (net +0.02) - a reminder it is not a stand-alone strategy - while Aluminium value stayed solidly positive (net +0.24). Its role is confirmed not by stand-alone Sharpe but by its diversification: it is the sleeve that turns contrarian when momentum is most exposed, which is exactly why it earns its third of the portfolio.

6. Combined Portfolio

6.1 Construction - Equal-Weight (default)

The three sleeves are combined daily into one position with a fixed one-third weight each:

Port_pos(t) = (1/3)·Momentum(t) + (1/3)·Carry(t) + (1/3)·Value(t)

Each leg is the metal-specific configuration selected for this Portfolio (Momentum and Value use each tab's best-performing signal; the Carry leg uses Z-score-252d rather than the Carry tab's own best performer, Carry-momentum-20d, for the taxonomy reason detailed in 4.3 and 6.6), all on Same-Day execution. The combined position ranges from -1 (all three short) to +1 (all three long); a value such as +0.33 means two sleeves long, one short. The weights are fixed - they do not move with market conditions - which keeps the portfolio simple and transparent.

6.2 Construction - Inverse-Volatility (alternative)

The dashboard also offers an inverse-volatility (risk-balanced) weighting. Equal-weight equalises position SIZE, not risk: an always-on sleeve (momentum) contributes more variance than one that is flat 40% of the time (value). Inverse-vol corrects this by giving each sleeve a weight inversely proportional to its own recent volatility, so each contributes roughly equal risk.

$\sigma_i(t)$ = trailing 63-day std. dev. of sleeve i's daily return, lagged one day

$$w_i(t) = (1 / \sigma_i(t)) / [1/\sigma_m(t) + 1/\sigma_c(t) + 1/\sigma_v(t)]$$

$$Port\_pos(t) = w_m(t)\cdot Momentum(t) + w_c(t)\cdot Carry(t) + w_v(t)\cdot Value(t)$$

- **Which volatility.** each sleeve's OWN realised daily strategy return (position × ΔF1_continuous / prior price), not raw price volatility. This measures how volatile that sleeve's P&L actually is.
- **Window.** a 63-trading-day (about three-month) trailing rolling standard deviation, lagged one day so today's weight uses only information available through yesterday - no look-ahead. (The 63-day window was itself chosen by a sweep: it is the Sharpe-optimal inverse-vol window; 10 days is too noisy, 126+ days gives up return for no Sharpe gain.)
- **Fixed or floating.** FLOATING, not fixed. The three weights are recomputed and renormalised to sum to 1 on every trading day, so the portfolio is rebalanced daily and tilts away from whichever sleeve is currently most volatile. During the first 63 days (volatility undefined) the weights fall back to 1/3 each.
- **Why it ends up near equal-weight. because the three sleeves have broadly similar realised volatility, 1/sigma is similar for all three, so the average weights land close to equal (the dashboard shows the live averages, e.g. about 0.26 / 0.26 / 0.39 for Copper - Value carries a larger share here because it turns over far less than the two trend/carry legs). Inverse-vol and equal-weight are close substitutes for Copper: full-sample Sharpe is close (0.78 vs 0.73, inverse-vol slightly ahead here), and inverse-vol has a materially shallower full-sample drawdown (-14% vs -25%); over the trailing five years the two converge further (net 0.91 vs 0.88, both around -12% drawdown). Equal-weight remains the default for simplicity.**

Every card, table and chart on the Portfolio tab recomputes for whichever weighting is selected; the figures in Section 6.4 below are the equal-weight default.

Legs by metal:

Sleeve	Copper	Aluminium
Momentum	MA(20,45)	MA(60,115)
Carry	Z-score 252-day	Z-score 252-day
Value	V1 F8, 5-year anchor	V1 F12, 5-year anchor

6.3 Why it works - low correlation

The diversification claim rests on the three position series being close to uncorrelated. Measured over the full sample:

Pair	Copper	Aluminium

Momentum - Carry	+0.13	+0.23
Momentum - Value	-0.21	-0.36
Carry - Value	-0.04	-0.01

Correlations are modest across the board (Copper below 0.25 in magnitude on all three pairs; Aluminium's momentum-value link is more meaningfully negative at -0.36, the strongest diversification pair in the framework). Combining three roughly independent sources of similar Sharpe lifts the portfolio Sharpe well above any single sleeve and roughly halves the drawdown.

6.4 Performance

Equal-weight portfolio, net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.76	0.73	+8.7%	-24%	18%
Copper - trailing 5y	0.92	0.88	+10.3%	-11%	9%
Aluminium - full sample	0.87	0.85	+9.3%	-21%	20%
Aluminium - trailing 5y	0.62	0.59	+6.2%	-15%	20%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.41 (Carry leg = Z-score-252d; +0.68 with the original MA(35,43)+CarryMom-20d configuration - see the leg-choice notes in 3.3, 4.3 and the research note in 6.6), Aluminium +1.12.

The portfolio dominates every individual sleeve on both metals and on both the full sample and the trailing five years. Its drawdown is markedly smaller than any individual sleeve (Copper -11% over the trailing five years versus -32% to -45% for the individual sleeves). The Aluminium walk-forward Sharpe of 1.12 - higher than its in-sample 0.85 - is strong evidence the combination is not over-fit.

6.5 Current positioning (latest available date)

Metal	Momentum	Carry	Value	EW Portfolio
Copper	+1 (long)	+1 (long)	-1 (short)	+0.33 (net long)
Aluminium	+1 (long)	+1 (long)	-1 (short)	+0.33 (net long)

Copper is net long (momentum and carry both bullish, value contrarian short); Aluminium is net long (trend and carry both bullish, value contrarian short). The split positioning across the two metals illustrates the framework operating independently per metal.

6.6 Research note: regime detection and Carry Momentum placement (not yet in production)

Two further research questions were investigated for the Portfolio but are not yet built into the live dashboard or reflected in the performance figures above.

- **Regime detection (R1/R2).** Two dynamic-weighting approaches were tested walk-forward on both metals: R1 (term-structure regime - tilt toward Momentum+Carry in backwardation, Value in contango, weights IS-optimised per window) and R2 (volatility regime - same structure, split on trailing realised vol). Neither reliably beat the static equal-weight portfolio on either metal: R1 averaged -0.06 to +0.06 OOS Sharpe on Copper depending on configuration, and was clearly negative on Aluminium (-0.16 avg, -0.20 post-2022, where backwardation is rare - only 15% of days - so the IS-fitted weights are starved of data and land on noisy corner solutions); R2 averaged -0.04 to +0.12 on Copper depending on the carry leg used, and -0.15 avg / -0.34 post-2022 on Aluminium - both metals negative post-2022 in every R2 configuration tested. Both are shelved - the static equal-weight blend keeps outperforming every dynamically re-weighted version tested, on both metals.

- **Carry Momentum placement.** The Portfolio's Carry leg is Z-score-252d rather than the tab's own best-performing signal, Carry-momentum-20d (Section 4.3), because Carry-momentum trades the CHANGE in the roll yield - structurally a momentum operation on a curve variable, which blurs the three-distinct-premia taxonomy the Portfolio is built on. This is supported by the literature: Boons and Prado (2019, Journal of Finance, "Basis-Momentum") identify momentum in the near-term futures basis as a return predictor distinct from both the basis level (carry) and price momentum - priced, maturity-specific, and increasing in volatility. A quick walk-forward test on Copper found Carry Momentum still adds real value when reintroduced deliberately rather than substituted for Carry: as an unconditional 4th equal-weight sleeve (average OOS Sharpe 0.54 vs the 3-sleeve base's 0.49), and more so as an overlay active only during backwardation (0.60) or high realised volatility (0.52) - the backwardation-conditional version actually beats the original all-carry-momentum 3-sleeve configuration (0.57). The same test on Aluminium shows the opposite for the unconditional version - adding Carry Momentum as a naive 4th sleeve HURTS (0.59 vs the 0.77 base), since Carry-momentum is already known to be weak/dead on Aluminium standalone - but the regime-conditional overlays are roughly neutral there (0.77-0.79), not harmful. This metal-specific asymmetry is a point in favour of the idea, not against it: the regime gating appears to be doing real work (excluding a bad signal on Aluminium) rather than being an artifact that only looks good on Copper. This is a promising Phase 2 candidate for Copper specifically - a 4th signal or regime-conditional overlay, not a leg substitution - but is not yet built into the live Portfolio tab or reflected in the figures above.

7. Cross-Metal Conclusions

- **The diversification thesis holds.** three economically distinct premia (trend, curve, mean-reversion) combine into a portfolio with materially higher Sharpe and roughly half the drawdown of any single sleeve, on both metals. This is the core of the framework.
- **Optima are metal-specific.** the same engine and the same three premia work on Aluminium, but the winning parameters differ - slower trend, z-score carry, F12 value. A one-size template would have understated Aluminium; per-metal selection is necessary.
- **Discipline matters.** removing the carry look-ahead, charging costs on the real traded price, and demanding walk-forward survival are what separate the real edges (trend, carry-momentum, z-score carry, V1 value) from the traps (level-carry over-statement, 1-day carry change, V2 reversal).
- **Value is a portfolio component, not a stand-alone.** value is weak alone, especially over the trailing five years, but earns its place through negative correlation to momentum. The portfolio, not the sleeve, is the product.

Caveats

- the per-metal best momentum pair and carry/value winners were chosen in-sample; their out-of-sample numbers hold up, but selection itself was not walk-forward.
- Value's full-sample edge is concentrated in the 2020-2021 dislocation.
- results are pre-financing, single-contract, ±1 unit sizing; no volatility targeting at the sleeve level.
- all figures reproduce live on the dashboard; the metal toggle on each of the four signal tabs switches the entire tab to that metal's configuration.
- open question for Hartree: two Portfolio leg choices were changed from the original grid-optimal configuration for trader-realism / taxonomy reasons rather than pure backtest performance - the Momentum leg (MA(35,43) to MA(20,45), see 3.3) and the Carry leg used inside the Portfolio only (Carry-momentum-20d to Z-score-252d, see 4.3 and 6.6). Combined, these cost real Sharpe versus the original all-grid-optimal configuration (EW net 0.73 vs 1.00 full-sample; walk-forward OOS +0.41 vs +0.68), while trailing-five-year performance is closer (net 0.88 vs 0.93). We would like Hartree's view on whether convention-realism and taxonomy cleanliness should outweigh a Sharpe cost of this size for a systematic implementation, or whether the grid-optimal parameters are preferable regardless of how a discretionary desk would frame them. The Carry Momentum research note (6.6) proposes a way to recover some of this cost without abandoning the cleaner taxonomy.
- open question for Hartree: whether the Anchors periods (MA(10,25)/(35,43)/(63,100)), or the single optimised MA(35,43), reflect spacing a trend desk would actually trade - a more textbook-style fast/medium/slow triad (MA(1,20)/(5,60)/(20,250)) tested negative out-of-sample under the same methodology, so convention and backtest robustness point in different directions here.

All performance figures verified against the live dashboard engine (PnL on F1_continuous, costs on F1_raw, active-day Sharpe, no look-ahead). Copper sample ends 2025-12-31; Aluminium sample ends 2026-05-19.